# MH1300 FOUNDATIONS OF MATHEMATICS

## Tutorial 1 Solutions

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**Ex. 1.1.7b, 7d.** Rewrite the following sentences less formally, without using variables. Determine, as best as you can, whether the statements are true or false.

b. There is a real number $x$ such that $x^2 < x$.

d. For all real numbers $a$ and $b$, $|a+b| \leq |a| + |b|$.

SOLUTION . b. There is a real number whose square is less than itself. Statement is true. E.g. take the real number $\frac{1}{2}$. Then we see that $\left(\frac{1}{2}\right)^2 = \frac{1}{4} < \frac{1}{2}$.

d. The absolute value of the sum of any two numbers is less than or equal to the sum of their absolute values. Statement is true. This is known as the triangle inequality, to be discussed in Section 4.4.

$\square$

**Ex. 1.1.9.** Fill in the blanks and rewrite the given statement.

SOLUTION . For all equations $E$, if $E$ is quadratic then $E$ has at most two real solutions.

a. All quadratic equations <u>have at most two real solutions</u>.

b. Every quadratic equation <u>has at most two real solutions</u>.

c. If an equation is quadratic, then it <u>has at most two real solutions</u>.

d. If $E$ <u>is a quadratic equation</u>, then $E$ <u>has at most two real solutions</u>.

e. For all quadratic equations $E$, <u>$E$ has at most two real solutions</u>.

$\square$

**Ex. 1.2.4.**

SOLUTION . a. Is $2 \in \{2\}$?
Ans: Yes.
b. How many elements are in the set $\{2, 2, 2, 2\}$?
Ans: 1.
c. How many elements are in the set $\{0, \{0\}\}$?
Ans: 2. The two elements are 0 and $\{0\}$
d. Is $\{0\} \in \{\{0\}, \{1\}\}$?
Ans: Yes.
e. Is $0 \in \{\{0\}, \{1\}\}$?
Ans: No. Because the set on the right has two elements, $\{0\}$ and $\{1\}$. Neither of these is equal to the element being asked. $\square$

**Ex. 1.2.7.** Use the set-roster notation to indicate the elements in each of the following sets.

a. $S = \{n \in \mathbb{Z} | \; n = (-1)^k, \text{ for some integer } k\}$.

b. $T = \{m \in \mathbb{Z} | \; m = 1 + (-1)^i, \text{ for some integer } i\}$.

c. $U = \{r \in \mathbb{Z} | \; 2 \leq r \leq -2\}$.

d. $V = \{s \in \mathbb{Z} | \; s > 2 \text{ or } s < 3\}$.

e. $W = \{t \in \mathbb{Z} | \; 1 < t < -3\}$.

f. $X = \{u \in \mathbb{Z} | \; u \leq 4 \text{ or } u \geq 1\}$.

SOLUTION . a. $S = \{-1, 1\}$.

b. $T = \{1 + (-1), 1 + 1\} = \{0, 2\}$.

c. $U = \{\}$. (Another notation is $\varnothing$.)

d. $V = \mathbb{Z} = \{\ldots, -2, -1, 0, 1, 2, \ldots\}$.

e. $W = \{\}$, or $W = \varnothing$.

f. $X = \mathbb{Z} = \{\ldots, -2, -1, 0, 1, 2, \ldots\}$.

□

**Ex. 1.2.9.**

SOLUTION .

| | | |
|---|---|---|
| a. | Is $3 \in \{1, 2, 3\}$? | Ans: Yes. |
| b. | Is $1 \subseteq \{1\}$? | Ans: No. |
| c. | Is $\{2\} \in \{1, 2\}$? | Ans: No. |
| d. | Is $\{3\} \in \{1, \{2\}, \{3\}\}$? | Ans: Yes. |
| e. | Is $1 \in \{1\}$? | Ans: Yes. |
| f. | Is $\{2\} \subseteq \{1, \{2\}, \{3\}\}$? | Ans: No. |
| g. | Is $\{1\} \subseteq \{1, 2\}$? | Ans: Yes. |
| h. | Is $1 \in \{\{1\}, 2\}$? | Ans: No. |
| i. | Is $\{1\} \subseteq \{1, \{2\}\}$? | Ans: Yes. |
| j. | Is $\{1\} \subseteq \{1\}$? | Ans: Yes. |

□

**Ex. 1.2.10b, 10d.**

b. Is $(5, -5) = (-5, 5)$?

d. Is $\left(\frac{-2}{-4}, (-2)^3\right) = \left(\frac{3}{6}, -8\right)$?

SOLUTION . b. No. The first coordinate on the left side is 5 and the first coordinate on the right side is -5, and $5 \neq -5$.

d. Yes. $\left(\frac{-2}{-4}, (-2)^3\right) = \left(\frac{1}{2}, -8\right) = \left(\frac{3}{6}, -8\right)$.

□

**Question E1.** Are the following pairs of sets equal? Explain your answer.

a. $\{0, 1, 3, 5\}$ and $\{n \in \mathbb{Z} \mid n^2 - 1 \leq n\}$

b. $\{1, 1, 3\}$ and $\{3, 3, 1\}$

c. $\{x \in \mathbb{R} \mid 0 < x \leq 2\}$ and $\{1, 2\}$

SOLUTION . a. The elements of the set $\{n \in \mathbb{Z} \mid n^2 - 1 \leq n\}$ are all the integers $n$ such that $n^2 - n - 1 \leq 0$. The roots of the equation $n^2 - n - 1$ are

$$n = \frac{1 \pm \sqrt{1+4}}{2} = \frac{1 \pm \sqrt{5}}{2} \approx 1.6, -0.6.$$

Thus the only integers $n$ such that $n^2 - n - 1 \leq 0$ are $n = 0, 1$. So the set $\{n \in \mathbb{Z} \mid n^2 - 1 \leq n\} = \{0, 1\}$. So the two sets are not equal.

Alternatively, since the question *did not* require us to list out the elements of the set $\{n \in \mathbb{Z} \mid n^2 - 1 \leq n\}$, we did not have to solve for the exact elements of the set as above. It is enough to observe that the number $n = 5$ does not satisfy the property $n^2 - 1 \leq n$ since $5^2 - 1 = 24 > 5$. So $5 \in \{0, 1, 3, 5\}$ but $5 \notin \{n \in \mathbb{Z} \mid n^2 - 1 \leq n\}$, and so the two sets are not equal.

b. The order of listing and repetition do not matter for sets. So, $\{1, 1, 3\} = \{1, 3\}$ and $\{3, 3, 1\} = \{3, 1\} = \{1, 3\}$. So they are equal.

c. $\{x \in \mathbb{R} \mid 0 < x \leq 2\}$ contains real numbers between 0 and 2, including 2. So for example, $\frac{1}{2} \in \{x \in \mathbb{R} \mid 0 < x \leq 2\}$ but $\frac{1}{2} \notin \{1, 2\}$. So they are not equal.

$\square$

**Question E2.** List out the elements of $B$ in set-roster notation. Is $A \subseteq B$?

$$A = \{1, 2, 3, 4\}, \qquad C = \{5, 6, 7, 8\}, \qquad B = \{n \in A \mid n + m = 8 \text{ for some } m \in C\}$$

SOLUTION . $B$ contains exactly the elements $n \in A = \{1, 2, 3, 4\}$ such that $n$ satisfies $n + m = 8$ for some $m \in \{5, 6, 7, 8\}$. So we need to go through each element of $A$ and see which ones have the property:

**$n = 1$:** We need to see if we can find some $m \in \{5, 6, 7, 8\}$ such that $1 + m = 8$. This is true if we take $m = 7$.

**$n = 2$:** We need to see if we can find some $m \in \{5, 6, 7, 8\}$ such that $2 + m = 8$. This is true if we take $m = 6$.

**$n = 3$:** We need to see if we can find some $m \in \{5, 6, 7, 8\}$ such that $3 + m = 8$. This is true if we take $m = 5$.

**$n = 4$:** We need to see if we can find some $m \in \{5, 6, 7, 8\}$ such that $4 + m = 8$. But we cannot find such an $m$, because no $m \in \{5, 6, 7, 8\}$ will work. This is because $4 + m \neq 8$ when we substitute $m = 5, 6, 7, 8$. So since no $m$ works, $n = 4$ is not in the set $B$.

Thus the set $B = \{1, 2, 3\}$ in set-roster notation. Clearly $A \nsubseteq B$ since $4 \in A$ but $4 \notin B$. $\square$

**Question E3.** Are the following true or false? Explain your answer.

a. $\{x\} \subseteq \{x\}$
b. $\{x\} \in \{x, \{x\}\}$
c. $\{x\} \in \{x\}$
d. $\{\{x\}\} \subsetneq \{x, \{x\}\}$

SOLUTION . a. True, because $LHS = \{x\}$ has only one element $x$, which is an element of $RHS$.

b. True, because the set $\{x, \{x\}\}$ has two elements $x$ and $\{x\}$.

c. False, because the set $\{x\}$ has only one element, $x$. So $\{x\}$ is not an element of the set $\{x\}$ because $\{x\} \neq x$.

d. To prove proper subset, we need to prove two things:
- $\{\{x\}\} \subseteq \{x, \{x\}\}$: This is true because $LHS$ has only the one element, $\{x\}$, while the $RHS$ has two elements $x$ and $\{x\}$.
- $\{\{x\}\} \neq \{x, \{x\}\}$: To do this we need to find an element of $RHS$ which is not an element of $LHS$. Obviously the element $x$ is such an element.

Thus $\{\{x\}\} \subsetneq \{x, \{x\}\}$.

$\square$

**Question E4.** Are the following true or false?

a. $\{x, y\} \subseteq \{x, \{y\}\}$
b. $\{x\} \in \{x, \{x, y\}\}$
c. $\{\} \in \{x\}$
d. $\{y, \{x\}, \{y\}\} \subsetneq \{y, \{x\}, y, \{y, y\}\}$

SOLUTION . a. False. The set on the left has two elements $x$ and $y$. The set on the right has two elements $x$ and $\{y\}$. The element $y$ does not appear as an element of the right set.

b. False. The set on the right has two elements, $x$ and $\{x, y\}$. Neither of these is equal to $\{x\}$.

c. False. The set on the right has only one element, $x$. This is not equal to $\{\}$.

d. False. The set on the left has three elements $y, \{x\}$ and $\{y\}$. The set on the right has also three elements $y, \{x\}$ and $\{y, y\} = \{y\}$. So the two sets are equal.

$\square$